

41.5 Multiple imputation

Filling a hole with a plausible value is not wrong; filling it with *one* plausible value and then forgetting that it was a guess is. Multiple imputation keeps the convenience of a complete rectangular data set and restores the uncertainty a single filling destroys, by filling several times and letting the variation between fillings speak.

41.5.1 The idea

Rubin's proposal (1978, 1987) has three parts.

1. **Impute.** Draw M independent completions $\mathbf{D}^{(1)}, \dots, \mathbf{D}^{(M)}$, each replacing $\mathbf{D}_{\text{mis}}$ by a draw from its posterior predictive distribution given $\mathbf{D}_{\text{obs}}$ under an **imputation model**.
2. **Analyse.** Apply the intended complete-data analysis to each completion, getting estimates $\hat{Q}_1, \dots, \hat{Q}_M$ and complete-data variance estimates $U_1, \dots, U_M$.
3. **Combine.** Pool them by [Theorem 41.5.1](#).

Step 1 is the only novelty. Drawing from the posterior predictive distribution means drawing the imputation model's *parameters* as well as the values; an imputation that conditions on a point estimate treats it as known, which is one of the ways the procedure fails.

Definition 41.5.1. Multiple imputation

Let $\boldsymbol{\gamma}$ parameterize a model for the conditional distribution of $\mathbf{D}_{\text{mis}}$ given $\mathbf{D}_{\text{obs}}$, with prior $p(\boldsymbol{\gamma})$. An imputation is **proper** in Rubin's sense if the M completions are independent draws from

$$f(\mathbf{D}_{\text{mis}} \mid \mathbf{D}_{\text{obs}}) = \int f(\mathbf{D}_{\text{mis}} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\gamma}) p(\boldsymbol{\gamma} \mid \mathbf{D}_{\text{obs}}) d\boldsymbol{\gamma}, \quad (41.5.1)$$

the posterior predictive distribution. Three standard constructions:

- (a) **Regression imputation with a parameter draw.** For a continuous variable modelled as normal given the others, draw $\sigma^{2*} = \text{SSE}/\chi_{k-q}^2$ from its posterior under a flat prior, then $\boldsymbol{\gamma}^* \sim N(\hat{\boldsymbol{\gamma}}, \sigma^{2*}(\mathbf{Z}_o^\top \mathbf{Z}_o)^{-1})$, then impute $\mathbf{z}^\top \boldsymbol{\gamma}^* + \sigma^* \varepsilon$ with ε standard normal. This is Theorem 7.6.2 with the noninformative limit Corollary 7.6.4.
- (b) **Predictive mean matching.** Compute the predicted value for the record with the gap, find the d recorded records with the closest predicted values, and impute one of their *observed* values at random. Every imputed value then actually occurred, so the marginal distribution is respected even when the normal model is wrong, at the cost of a less transparent theory.
- (c) **Chained equations.** With gaps in several variables, cycle through them, imputing each in turn from a regression on the current versions of all the others, for several sweeps, and keep the completion at the end of a sweep.

Construction (c), also called fully conditional specification or MICE (van Buuren, 2018), is what makes multiple imputation usable on real files, where dozens of variables have gaps and no one will write a joint density for all of them. It is a Gibbs-like sweep (compare Example (The proficiency study, four ways)) carrying a risk the Gibbs sampler does not: the specified conditionals need not correspond to *any* joint distribution, and then the sweep has no stationary distribution and the result depends on the order of the variables. The dependence is usually

slight, and the flexibility is worth the untidiness, but it should be known.

41.5.2 Rubin's rules

Theorem 41.5.1. Combining M analyses

Let Q be the estimand and suppose the complete-data analysis is calibrated in the sense that, given the complete data $\mathbf{D}$, it reports $\hat{Q}(\mathbf{D})$ and $U(\mathbf{D})$ with

$$E(Q \mid \mathbf{D}) = \hat{Q}(\mathbf{D}), \quad \text{Var}(Q \mid \mathbf{D}) = U(\mathbf{D}). \quad (41.5.2)$$

Let $\mathbf{D}^{(1)}, \dots, \mathbf{D}^{(M)}$ be independent draws from [Eq. 41.5.1](#), generated by the analyst and therefore conditionally independent of Q given $\mathbf{D}_{\text{obs}}$; put $\hat{Q}_\ell = \hat{Q}(\mathbf{D}^{(\ell)})$ and $U_\ell = U(\mathbf{D}^{(\ell)})$, and define

$$\bar{Q} = \frac{1}{M} \sum_{\ell} \hat{Q}_\ell, \quad \bar{U} = \frac{1}{M} \sum_{\ell} U_\ell, \quad B = \frac{1}{M-1} \sum_{\ell} (\hat{Q}_\ell - \bar{Q})^2. \quad (41.5.3)$$

Then, with all expectations conditional on $\mathbf{D}_{\text{obs}}$:

- (a) (*Decomposition.*) $\text{Var}(Q \mid \mathbf{D}_{\text{obs}}) = E(U \mid \mathbf{D}_{\text{obs}}) + \text{Var}(\hat{Q} \mid \mathbf{D}_{\text{obs}})$, and $\bar{U}$ and B are unbiased for the two terms;
- (b) (*Total variance.*) $\bar{Q}$ is unbiased for $E(Q \mid \mathbf{D}_{\text{obs}})$, and

$$T = \bar{U} + \left(1 + \frac{1}{M}\right)B \quad (41.5.4)$$

is unbiased for the variance of $Q - \bar{Q}$ given $\mathbf{D}_{\text{obs}}$;

- (c) (*Degrees of freedom.*) treating $\bar{U}$ as fixed and $(M-1)B / \text{Var}(\hat{Q} \mid \mathbf{D}_{\text{obs}})$ as χ_{M-1}^2 , the variance of T matches that of a scaled χ_ν^2 variable with

$$\nu = (M-1) \left\{ \frac{T}{(1 + M^{-1})B} \right\}^2 = \frac{M-1}{\gamma^2}, \quad \gamma = \frac{(1 + M^{-1})B}{T}, \quad (41.5.5)$$

and the interval $\bar{Q} \pm t_{\nu, 1-\alpha/2} \sqrt{T}$ is the standard multiple-imputation interval; γ is the estimated **fraction of missing information**;

- (d) (*Efficiency.*) write $T_\infty = E(U) + \text{Var}(\hat{Q})$ for the total variance at $M = \infty$ and $\gamma_\infty = \text{Var}(\hat{Q})/T_\infty$ for the limiting fraction of missing information. Then the target of T exceeds T_∞ by the factor $1 + \gamma_\infty/M$, so using M imputations rather than infinitely many inflates the standard error by $(1 + \gamma_\infty/M)^{1/2}$.

Proof. Throughout, condition on $\mathbf{D}_{\text{obs}}$ and write E and Var for the conditional operators.

- (a) The conditional variance decomposition applied to Q with the intermediate conditioning variable $\mathbf{D}$ gives $\text{Var}(Q) = E\{\text{Var}(Q \mid \mathbf{D})\} + \text{Var}\{E(Q \mid \mathbf{D})\}$, which is $E(U) + \text{Var}(\hat{Q})$ by [Eq. 41.5.2](#). The completions are draws of $\mathbf{D}$ from its conditional law, so $E\bar{U} = E(U)$, and the sample variance of independent draws is unbiased for their variance, so $E B = \text{Var}(\hat{Q})$.
- (b) $E\bar{Q} = E\hat{Q} = E\{E(Q \mid \mathbf{D})\} = E Q$, so $\bar{Q}$ is unbiased for the posterior mean. Because

the completions are conditionally independent of Q given $\mathbf{D}_{\text{obs}}$, so is $\bar{Q}$, and therefore

$$\begin{aligned}\text{Var}(Q - \bar{Q}) &= \text{Var}(Q) + \text{Var}(\bar{Q}) = \{E(U) + \text{Var}(\hat{Q})\} + \frac{\text{Var}(\hat{Q})}{M} \\ &= E(U) + \left(1 + \frac{1}{M}\right) \text{Var}(\hat{Q}),\end{aligned}$$

using (a) for the first bracket and independence of the M draws for the second term. By (a) again, $\bar{U}$ and $(1 + M^{-1})B$ are unbiased for the two pieces, so T is unbiased for the whole.

- (c) Under the stated approximation $T = \bar{U} + (1 + M^{-1})B$ has variance $(1 + M^{-1})^2 \text{Var}(B)$, which is $2(1 + M^{-1})^2 \{\text{Var}(\hat{Q})\}^2 / (M - 1)$. A scaled χ^2_ν variable with mean T has variance $2T^2/\nu$. Equating and replacing $\text{Var}(\hat{Q})$ by its estimate B ,

$$\frac{2T^2}{\nu} = \frac{2(1 + M^{-1})^2 B^2}{M - 1} \implies \nu = (M - 1) \frac{T^2}{(1 + M^{-1})^2 B^2},$$

which is [Eq. 41.5.5](#).

- (d) By (b) the target of T is

$$E(U) + (1 + M^{-1}) \text{Var}(\hat{Q}) = T_\infty + \frac{\text{Var}(\hat{Q})}{M} = T_\infty \left(1 + \frac{\gamma_\infty}{M}\right).$$

Taking square roots gives the inflation factor.

■

Remark (What the proof assumes). The calibration [Eq. 41.5.2](#) is a Bayesian statement: the complete-data analysis behaves as a posterior mean and variance would. Rubin's rules are therefore exactly right when imputation and analysis come from the *same* joint model, and approximately right otherwise. The degrees of freedom in part (c) can exceed the complete-data degrees of freedom, absurdly, when M is large and the sample small; Barnard and Rubin (1999) give the corrected formula.

Idea. The between-imputation variance B is the price of not knowing the missing values, and $\bar{U}$ is the price of having a finite sample. Adding them is the whole of multiple imputation. The factor $1 + 1/M$ is the price of using finitely many imputations to measure B .

41.5.3 How many imputations

Part (d) is why the classical advice was “three to five imputations”. With $\gamma_\infty = 0.3$ and $M = 5$ the standard error is inflated by 1.0296, three per cent; with $\gamma_\infty = 0.5$ the inflation is 1.0488 at $M = 5$ and 1.0124 at $M = 20$.

The advice has shifted upward since, for two reasons the theorem does not address. The degrees of freedom [Eq. 41.5.5](#) grow like $M - 1$, so a small M gives a fat-tailed reference distribution and wide intervals even when the point estimate is fine; and $\bar{Q}$ is itself a Monte Carlo average, so a second analyst with a different seed gets a different number — uncomfortable when that number is a p -value near a threshold. Imputations are cheap, and the curve in [Figure 41.5.1\(b\)](#)

is flat past $M = 20$ for every plausible γ .

41.5.4 Coverage in practice

Example. Five analyses of one problem

Take $n = 200$ records with two correlated covariates and $y = 1 + 0.5x_1 + 0.6x_2 + \varepsilon$, and lose x_2 with a probability depending on x_1 and y — both recorded, so MAR — leaving about 106 complete records. Over 2000 replicates, for the coefficient of x_2 :

Table 41.5.1. Five analyses of the same MAR problem. The estimand is $\beta_2 = 0.6$; “mean se” is the average reported standard error and “sd” the actual spread of the estimates over 2000 replicates.

analysis	bias	mean se	sd	coverage
complete cases	−0.151	0.101	0.102	67.2%
mean imputation	−0.319	0.125	0.108	25.6%
single imputation	−0.010	0.082	0.138	75.9%
multiple imputation, $M = 5$	−0.013	0.118	0.119	95.0%
multiple imputation, $M = 20$	−0.013	0.115	0.114	94.5%

Each row fails in its own way. The complete cases are biased, the missingness depending on the response. Mean imputation is far worse, and its reported standard error is *larger* than its actual spread. A single proper imputation removes the bias but reports a standard error three fifths of the truth, and covers 75.9% of the time. Multiple imputation with five draws gets both right, with an average estimated fraction of missing information 0.463 and median degrees of freedom 18.3.

```
import numpy as np

n = 200
rho = 0.5
beta_true = np.array([1.0, 0.5, 0.6])           # intercept, x1, x2

def make_data(rng):
    x1 = rng.normal(size=n)
    x2 = rho * x1 + np.sqrt(1 - rho**2) * rng.normal(size=n)
    y = beta_true[0] + beta_true[1] * x1 + beta_true[2] * x2 + rng.normal(size=n)
    eta = -0.3 + 0.8 * x1 + 2.0 * (y - y.mean())    # depends on observed values
    # only: MAR
    seen = rng.uniform(size=n) > 1 / (1 + np.exp(-eta))
    return x1, x2, y, seen
```

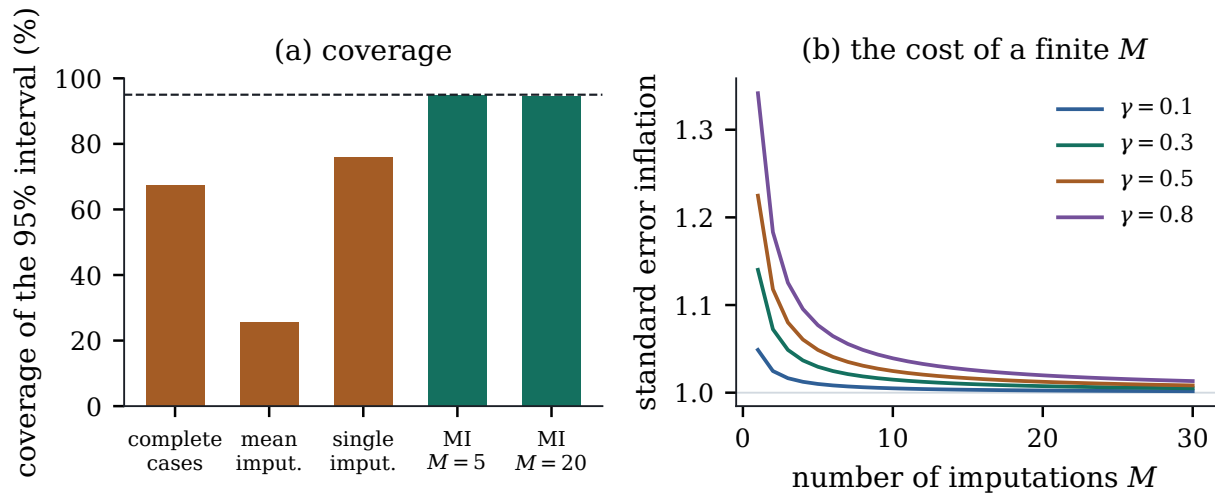


Figure 41.5.1: (a) Coverage of nominal 95% intervals for β_2 under five analyses of the same missing-at-random problem. (b) The factor by which a finite number of imputations inflates the standard error, as a function of M , for four values of the fraction of missing information.

```
def fit(X, y):
    """Least squares with the coefficient of x2 and its squared standard error."""
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    dof = len(y) - X.shape[1]
    s2 = np.sum((y - X @ b) ** 2) / dof
    v = s2 * np.linalg.inv(X.T @ X)[2, 2]
    return b[2], v, dof

def impute_once(x1, x2, y, seen, rng):
    """One draw from the posterior predictive distribution of the missing x2."""
    Z = np.column_stack([np.ones(n), x1, y]) # imputation model: x2 on x1
    # and y
    Zo, k = Z[seen], int(seen.sum())
    g, *_ = np.linalg.lstsq(Zo, x2[seen], rcond=None)
    sse = float(np.sum((x2[seen] - Zo @ g) ** 2))
    s2_star = sse / rng.chisquare(k - Z.shape[1]) # draw the variance ...
    cov = s2_star * np.linalg.inv(Zo.T @ Zo)
    g_star = rng.multivariate_normal(g, cov) # ... then the coefficients
    draw = Z @ g_star + np.sqrt(s2_star) * rng.normal(size=n)
    return np.where(seen, x2, draw) # observed values are kept

def rubin(estimates, variances):
    """Combine M analyses: the average, the total variance and the degrees of
    freedom."""
    M = len(estimates)
    q_bar, u_bar = np.mean(estimates), np.mean(variances)
    b = np.var(estimates, ddof=1)
    total = u_bar + (1 + 1 / M) * b
    gamma = (1 + 1 / M) * b / total # fraction of missing
    # information
    nu = (M - 1) / gamma**2
    return q_bar, total, nu

demo_rng = np.random.default_rng(4105)
```

```

x1d, x2d, yd, seend = make_data(demo_rng)           # one data set, x2 missing at
random
qs, us = [], []
for _ in range(5):                                # M = 5 completions of it
    xi = impute_once(x1d, x2d, yd, seend, demo_rng)
    qi, ui, _ = fit(np.column_stack([np.ones(n), x1d, xi]), yd)
    qs.append(qi)
    us.append(ui)
q_bar, total, nu = rubin(qs, us)
print(f"M = 5: estimate {q_bar:.3f}, total variance {total:.4f}, "
      f"degrees of freedom {nu:.1f}")

```

[Run this code online](#)

Chained equations handle gaps in more than one variable: ten sweeps of the two conditional regressions below recover the coefficient of x_2 when both covariates are incomplete.

```

rho = 0.5
beta_true = np.array([1.0, 0.5, 0.6])              # intercept, x1, x2

def make_data(rng):
    x1 = rng.normal(size=n)
    x2 = rho * x1 + np.sqrt(1 - rho**2) * rng.normal(size=n)
    y = beta_true[0] + beta_true[1] * x1 + beta_true[2] * x2 + rng.normal(size=n)
    eta = -0.3 + 0.8 * x1 + 2.0 * (y - y.mean())    # depends on observed values
    only: MAR
    seen = rng.uniform(size=n) > 1 / (1 + np.exp(-eta))
    return x1, x2, y, seen

rng2 = np.random.default_rng(4107)
x1, x2, y, seen2 = make_data(rng2)
seen1 = rng2.uniform(size=n) > 0.25                # x1 now has gaps of its own
x1_work = np.where(seen1, x1, x1[seen1].mean())    # a crude start
x2_work = np.where(seen2, x2, x2[seen2].mean())

for sweep in range(10):                            # one sweep = one pass over
    the variables
    Z = np.column_stack([np.ones(n), x2_work, y])    # x1 given the others
    g, *_ = np.linalg.lstsq(Z[seen1], x1[seen1], rcond=None)
    s = np.std(x1[seen1] - Z[seen1] @ g, ddof=3)
    x1_work = np.where(seen1, x1, Z @ g + s * rng2.normal(size=n))
    Z = np.column_stack([np.ones(n), x1_work, y])    # x2 given the others
    g, *_ = np.linalg.lstsq(Z[seen2], x2[seen2], rcond=None)
    s = np.std(x2[seen2] - Z[seen2] @ g, ddof=3)
    x2_work = np.where(seen2, x2, Z @ g + s * rng2.normal(size=n))
b_chained, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x1_work, x2_work]), y,
    rcond=None)
print("after chained equations:", np.round(b_chained, 3))

```

[Run this code online](#)

41.5.5 Congeniality

Multiple imputation splits the work between two models: the imputer's and the analyst's. Nothing forces them to agree, and Meng (1994) showed that the consequences depend on which

way they disagree.

The imputation model omits something the analysis uses. If the analyst fits an interaction and the imputer imputed one of its covariates from a model without it, the imputed values carry no interaction and the analyst's interaction is biased towards zero. This is the serious failure, and easy to commit: the imputer usually works first, for a whole file, without knowing which analyses will be run.

The imputation model includes something the analysis does not. Then the imputations are richer than the analysis needs; estimates are usually fine and Rubin's T tends to be *too large*, so the procedure is conservative rather than wrong.

Meng calls a pair of models **congenial** when both are implied by a single Bayesian model, and shows that Rubin's variance estimator is consistent under congeniality and can be biased either way without it. Two rules follow: put into the imputation model every variable the analysis will use, including the response and every transformation and interaction to be fitted; and when imputer and analyst are different people, err towards a rich imputation model.

Warning. Omitting the response from the imputation model is the single commonest mistake. It looks cautious — “I do not want to build the answer into the data” — and it is exactly backwards: it imputes covariates as though they were unrelated to the response, which biases every coefficient towards zero. The response belongs in the imputation model.

41.5.6 Exercises

41.5.6.1 A. Check your understanding

Exercise. A1

Set $M = 1$ in [Theorem 41.5.1](#) and explain why the formulas break down. Which of the three quantities in [Eq. 41.5.3](#) cannot be computed, and what does that say about single imputation?

Solution. With $M = 1$ the between-imputation sum of squares has no degrees of freedom: B is $0/0$. So the part of the uncertainty that comes from not knowing the missing values cannot be estimated at all from one imputation, which is exactly why a single imputation reports $\bar{U}$ alone and understates the variance.

Exercise. A2

From [Eq. 41.5.5](#), compute ν for $M = 5$ and $\gamma \in \{0.1, 0.3, 0.5\}$. At which value of γ does the multiple-imputation reference distribution become so heavy-tailed that the interval is more than 20% wider than a normal-based one?

Solution. Equation [Eq. 41.5.5](#) gives $\nu = (M - 1)/\gamma^2$, so at $M = 5$ the three values are $4/0.01 = 400$, $4/0.09 = 44.4$ and $4/0.25 = 16$. The interval is wider than the normal-based one by the ratio of the t quantile to 1.96, and a table shows that ratio passing 1.2 at about

seven degrees of freedom, that is at $\gamma = \sqrt{4/7.2} \approx 0.75$. Only a problem in which three quarters of the information is missing costs that much at $M = 5$ — and raising M is the cheap repair.

41.5.6.2 B. Practice

Exercise. B1

Construct a simulation in which the analysis fits y on x_1 , x_2 and their interaction, while x_2 is imputed from a model without the interaction. Measure the bias in the interaction coefficient and its coverage. Then repeat with the interaction included in the imputation model.

41.5.6.3 C. Going deeper

Exercise. C1

Show that when the imputation model and the analysis model are the same Bayesian model, $\bar{Q}$ and T converge as $M \rightarrow \infty$ to the exact posterior mean and variance of Q given $\mathbf{D}_{\text{obs}}$. Deduce that multiple imputation is, in the congenial case, a Monte Carlo scheme for a posterior that could also be sampled directly — and say why one might still prefer the imputation route.